

NASSAU CANDY FACTORY OPTIMIZATION

Factory Reallocation & Shipping Optimization Recommendation System

Research Paper • Data Analytics, Predictive Modeling & Decision Intelligence

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Project Type	Predictive Analytics & Decision Support
Dataset	Nassau Candy Distributor
Dataset Size	10,194 validated observations × 18 fields
Study Scope	Factory allocation, shipping lead time, route performance and scenario optimization
Delivery Layer	Interactive Streamlit decision-support dashboard

Decision objective: improve shipping efficiency while avoiding unsupported factory reallocations that could create operational or profit risk.

Executive Summary

This project develops a decision-intelligence system for Nassau Candy Distributor to evaluate factory assignments and simulate alternative factory configurations before operational execution. The objective is to move beyond descriptive reporting toward predictive and scenario-based decision support.

The validated dataset contains 10,194 observations and 18 fields covering order dates, ship dates, ship mode, customer geography, product, sales, units, gross profit and cost. Data validation found no duplicate rows, no invalid dates, no negative lead times, no non-positive sales or units, no negative cost, and no gross-profit calculation mismatches.

The analytical workflow combines feature engineering, predictive modeling, route clustering, factory-distance analysis and a scenario simulation engine. Three regression approaches were evaluated: Linear Regression, Random Forest Regressor and Gradient Boosting Regressor. Gradient Boosting was selected because it achieved the lowest RMSE among the tested models and the highest R^2 .

The selected model achieved RMSE 181.58, MAE 179.96 and R^2 0.5338 on the evaluation data. The optimization engine evaluated 120 product-region-factory scenarios. No alternative factory assignment produced a predicted lead-time improvement, resulting in 24 'Keep Current Factory' decisions for the reliable product-region combinations.

The result is intentionally conservative: the system does not force a positive recommendation when the model does not support one. Because factory-specific operating-cost data are unavailable, profit impact is treated as a stability constraint rather than a claimed factory-level profit improvement.



[Dashboard Overview]

1. Introduction

1.1 Business Context

Factory assignment is an operational decision that can affect shipping lead time, geographic efficiency and the economics of fulfilling customer demand. Static assignment rules may work for stable demand patterns, but they can become inefficient when product demand and destination patterns vary.

A decision-support system can improve this process by estimating expected shipping outcomes under alternative factory configurations and ranking the alternatives using measurable operational criteria.

1.2 Problem Statement

Nassau Candy Distributor currently uses product-to-factory assignments that can be difficult to evaluate before execution. The organization lacks a systematic mechanism to simulate alternative factory assignments, quantify predicted lead-time changes and compare operational trade-offs at scale.

- Potentially suboptimal shipping distances for some product-region combinations.
- Limited ability to quantify the effect of moving a product to another factory before execution.
- No unified scenario engine for comparing all available factories.
- Limited decision support for balancing shipping speed, geographic efficiency and profit stability.

1.3 Objectives

- Validate and prepare the Nassau Candy Distributor dataset for analysis.
- Engineer operational features relevant to shipping lead time.
- Predict expected shipping lead time using multiple regression models.
- Identify route and product-region performance patterns through clustering.
- Simulate alternative factory assignments for reliable product-region combinations.
- Rank factory scenarios using lead-time, distance and profit-stability signals.
- Provide actionable recommendations through an interactive Streamlit dashboard.

2. Research Questions

- Can shipping lead time be predicted from product, factory, region, ship mode and temporal features?
- Which regression model provides the strongest predictive performance for the available data?
- Which product-region combinations show relatively slower observed lead times?
- How do alternative factory assignments change predicted lead time?
- Do any alternative factory assignments provide a predicted operational improvement?
- How can geographic distance and profit stability be incorporated into recommendation ranking?
- How can the final analytical results be presented as an interactive decision-support tool?

3. Dataset and Data Quality

The project uses the supplied Nassau Candy Distributor dataset. Each row represents an order-level business observation containing product, customer geography, shipping, sales and profitability information.

Field	Type	Purpose
Order ID	Categorical	Order identification
Order Date / Ship Date	Date	Lead-time calculation and temporal features
Ship Mode	Categorical	Shipping-service context
Country/Region, City, State/Province, Postal Code	Categorical	Customer geography
Division / Region	Categorical	Business and destination grouping
Product ID / Product Name	Categorical	Product-level analysis
Sales	Numeric	Revenue measure
Units	Numeric	Order quantity
Gross Profit	Numeric	Profitability measure
Cost	Numeric	Cost measure

3.1 Data Validation Results

Validation Check	Result
Dataset shape	10,194 rows × 18 columns
Duplicate rows	0
Invalid order dates	0
Invalid ship dates	0
Negative lead times	0
Sales <= 0	0
Units <= 0	0
Negative cost	0
Gross-profit calculation mismatches	0

The dataset covers Order Dates from 2 January 2024 to 31 December 2025 and Ship Dates from 30 June 2026 to 28 June 2030. The resulting lead-time values are unusually large. These values were retained because they are present in the supplied data rather than silently corrected.

```

# 1. Check duplicate rows
print("Duplicate rows:", df.duplicated().sum())

# 2. Convert dates
df["Order Date"] = pd.to_datetime(
    df["Order Date"],
    format="%d-%m-%Y",
    errors="coerce"
)

df["Ship Date"] = pd.to_datetime(
    df["Ship Date"],
    format="%d-%m-%Y",
    errors="coerce"
)

# 3. Check invalid dates
print("Invalid Order Dates:", df["Order Date"].isna().sum())
print("Invalid Ship Dates:", df["Ship Date"].isna().sum())

# 4. Create shipping lead time
df["Lead Time"] = (
    df["Ship Date"] - df["Order Date"]
).dt.days

print("\nLead Time Summary:")
print(df["Lead Time"].describe())

# 5. Check impossible lead times
print("Negative Lead Times:", df["Lead Time"] < 0).sum())

# 6. Check financial values
print("Invalid Sales < 0:", (df["Sales"] < 0).sum())
print("Invalid Units < 0:", (df["Units"] < 0).sum())
print("Invalid Cost < 0:", (df["Cost"] < 0).sum())

# 7. Check profit calculation
profit_difference = (
    df["Sales"] - df["Cost"] - df["Gross Profit"]
).abs()

print(
    "\nGross Profit calculation mismatches:",
    (profit_difference > 0.00).sum()
)

print("\nData validation completed.")
✓ 0.0s

```

Duplicate rows: 0
 Invalid Order Dates: 0
 Invalid Ship Dates: 0

Lead Time Summary:
 count 10194.000000
 mean 1320.841868
 std 262.444892
 min 904.000000
 25% 1271.000000
 50% 1274.000000
 75% 1638.000000
 max 1642.000000
 Name: Lead Time, dtype: float64

Negative Lead Times: 0

Sales <= 0: 0
 Units <= 0: 0
 Cost < 0: 0

Gross Profit calculation mismatches: 0

[Data Validation Output]

4. Analytical Methodology

4.1 Data Preparation and Feature Engineering

Order Date and Ship Date were converted to datetime values. Shipping lead time was calculated as the difference between Ship Date and Order Date. Additional temporal and unit-economics features were created.

Feature	Definition	Purpose
Lead Time	Ship Date – Order Date	Target for predictive modeling
Order Year	Year of Order Date	Temporal pattern
Order Month	Month of Order Date	Seasonal context
Order Quarter	Quarter of Order Date	Quarterly context
Profit Margin	Gross Profit / Sales × 100	Profitability monitoring
Sales per Unit	Sales / Units	Unit economics
Cost per Unit	Cost / Units	Unit cost context

4.2 Factory Assignment Mapping

The dataset did not contain an explicit origin-factory field. A product-to-factory mapping supplied as part of the project methodology was therefore used to infer the current factory assignment for each product.

Factory	Mapped Products
Lot's O' Nuts	Nutty Crunch Surprise; Fudge Mallows; Scrumdiddlyumptious
Wicked Choccy's	Milk Chocolate; Triple Dazzle Caramel
Sugar Shack	Laffy Taffy; SweeTARTS; Nerds; Fun Dip; Fizzy Lifting Drinks
Secret Factory	Everlasting Gobstopper; Lickable Wallpaper; Wonka Gum
The Other Factory	Hair Toffee; Kazookles

4.3 Predictive Modeling

The predictive task estimates shipping lead time using Product Name, Current Factory, Region, Ship Mode, Order Year, Order Month and factory-to-region geographic distance. Three regression models were evaluated.

Model	RMSE	MAE	R ²
Linear Regression	182.43	181.41	0.5294
Random Forest Regressor	192.47	172.88	0.4762
Gradient Boosting Regressor	181.58	179.96	0.5338

Gradient Boosting was selected as the primary scenario-prediction model because it achieved the lowest RMSE among the tested models and the highest R². Random Forest achieved a slightly lower MAE, so model selection considers both predictive accuracy and interpretability rather than relying on a single metric.

```

from sklearn.pipeline import Pipeline
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

# Model features
model_data = df[
    [
        "Product Name",
        "Current Factory",
        "Region",
        "Ship Mode",
        "Order Year",
        "Order Month",
        "Factory_Region_Distance_km",
        "Lead Time"
    ]
].copy()

X = model_data.drop(columns=["Lead Time"])
y = model_data["Lead Time"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)

categorical_features = [
    "Product Name",
    "Current Factory",
    "Region",
    "Ship Mode"
]

numerical_features = [
    "Order Year",
    "Order Month",
    "Factory_Region_Distance_km"
]

preprocessor = ColumnTransformer(
    transformers=[
        (
            "num",
            StandardScaler(),
            numerical_features
        ),
        (
            "cat",
            OneHotEncoder(
                handle_unknown="ignore",
                sparse_output=False
            ),
            categorical_features
        )
    ]
)

gradient_boosting_model = Pipeline(
    steps=[
        ("preprocessor", preprocessor),
        ("model",
         GradientBoostingRegressor(
             random_state=42
         )
        )
    ]
)

gradient_boosting_model.fit(X_train, y_train)

print("Gradient Boosting model trained successfully!")

```

[18] ✓ 0.9s

... Gradient Boosting model trained successfully!

[Model Comparison]

4.4 Route Clustering

Reliable product-region combinations were defined using a minimum order-count threshold of 30. This produced 24 reliable routes from 53 observed product-region combinations. K-means clustering was evaluated using silhouette scores.

Number of Clusters (k)	Silhouette Score
2	0.5568
3	0.4319
4	0.3630
5	0.3942

Two clusters were selected because k=2 produced the highest silhouette score of 0.5568.

4.5 Scenario Simulation

For each reliable product-region combination, the system evaluated all five available factories. The resulting 24 × 5 design produced 120 scenarios. Each scenario contains predicted lead time, lead-time improvement, geographic distance, profit-margin stability and a recommendation score.

4.6 Recommendation Logic

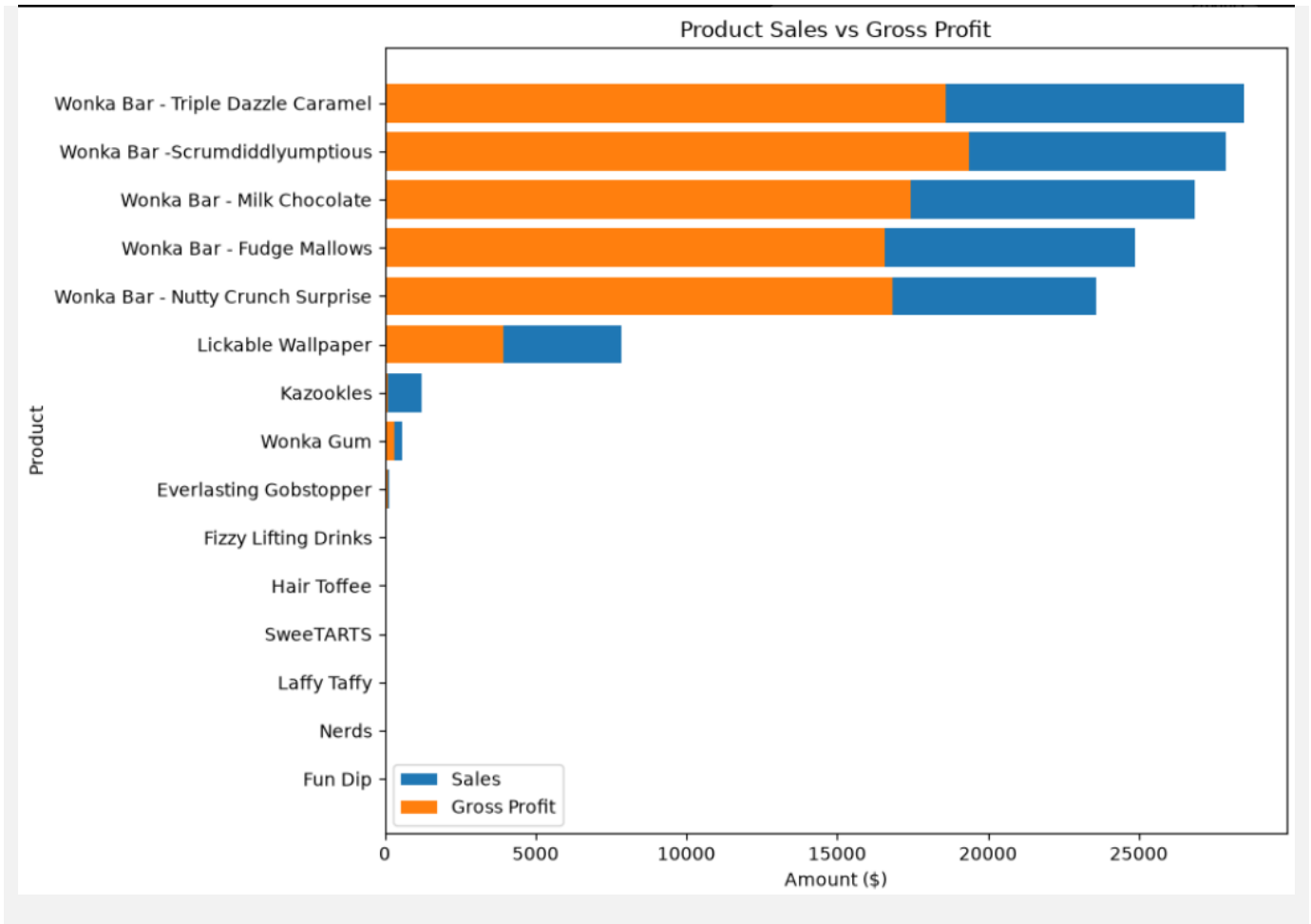
- Lead-time reduction is treated as the primary operational improvement signal.
- Geographic distance contributes to scenario ranking through a normalized distance score.
- Profit stability is treated as a constraint because factory-specific operating-cost data are unavailable.
- Scenarios with positive predicted lead-time improvement would be candidates for reassignment.
- If no alternative scenario improves lead time, the system retains the current factory.

5. Exploratory Data Analysis

5.1 Product Performance

Chocolate products dominate the dataset, accounting for 9,844 of the 10,194 observations. The highest-volume products are the major Wonka Bar variants. This imbalance means recommendations should prioritize high-volume products and avoid allowing very small samples to dominate operational decisions.

Product	Orders	Sales	Gross Profit	Margin
Triple Dazzle Caramel	1,677	\$28,485.00	\$18,610.20	65.33%
Scrumdiddlyumptious	1,704	\$27,874.80	\$19,357.50	69.44%
Milk Chocolate	1,768	\$26,867.75	\$17,443.37	64.92%
Fudge Mallows	1,527	\$24,890.40	\$16,593.60	66.67%
Nutty Crunch Surprise	1,529	\$23,574.95	\$16,819.95	71.35%



[Product Sales vs Gross Profit]

5.2 Regional Performance

Region	Orders	Sales	Gross Profit	Avg Lead Time
Interior	1,979	\$32,037.60	\$21,282.49	1,323.09
Atlantic	2,476	\$41,197.24	\$26,973.70	1,322.75
Pacific	2,744	\$46,301.53	\$30,485.94	1,322.19
Gulf	1,350	\$22,247.26	\$14,700.67	1,311.37

Regional average lead times are relatively close, indicating that region alone is not sufficient to explain shipping performance. Product, factory assignment and temporal effects therefore remain important.

5.3 Ship Mode Performance

Ship Mode	Orders	Avg Lead Time	Margin
Standard Class	5,142	1,314.33	66.00%
Second Class	1,653	1,323.85	65.71%
Same Day	442	1,333.44	66.08%
First Class	1,312	1,338.28	65.72%

5.4 Product-Region Route Analysis

Among reliable product-region combinations, Nutty Crunch Surprise in the Atlantic region stands out with 410 orders and an observed average lead time of 1,360.43. Other high-volume routes show smaller differences. This supports the use of product-region combinations rather than region-only analysis.

```

# Total product-region recommendations: 24



|    | Product Name                      | Region   | Current Factory | Scenario Factory  | Current Avg Lead Time | Predicted Lead Time | Lead Time Reduction % | Distance km | Profit Stability Score | Recommendation Score | Recommendation       |
|----|-----------------------------------|----------|-----------------|-------------------|-----------------------|---------------------|-----------------------|-------------|------------------------|----------------------|----------------------|
| 1  | Lickable Wallpaper                | Atlantic | Secret Factory  | Wicked Choccy's   | 1387.828571           | 1498.177227         | -7.95                 | 946.14      | 1                      | 33.054982            | Keep Current Factory |
| 9  | Wonka Bar - Milk Chocolate        | Interior | Wicked Choccy's | Secret Factory    | 1331.977956           | 1457.382668         | -9.41                 | 845.39      | 1                      | 32.860564            | Keep Current Factory |
| 11 | Wonka Bar - Nutty Crunch Surprise | Atlantic | Lot's O' Nuts   | Wicked Choccy's   | 1360.426531           | 1474.595636         | -8.39                 | 946.14      | 1                      | 32.780962            | Keep Current Factory |
| 13 | Wonka Bar - Nutty Crunch Surprise | Interior | Lot's O' Nuts   | Secret Factory    | 1329.847345           | 1457.382668         | -8.59                 | 845.39      | 1                      | 32.752564            | Keep Current Factory |
| 21 | Wonka Bar - Scrumdiddlyumptious   | Interior | Lot's O' Nuts   | Secret Factory    | 1229.908487           | 1459.347515         | -8.73                 | 845.39      | 1                      | 32.668564            | Keep Current Factory |
| 17 | Wonka Bar - Triple Dazzle Caramel | Interior | Wicked Choccy's | Secret Factory    | 1316.334812           | 1457.790127         | -10.75                | 845.39      | 1                      | 32.066564            | Keep Current Factory |
| 5  | Wonka Bar - Fudge MalloWS         | Interior | Lot's O' Nuts   | Secret Factory    | 1311.570732           | 1457.679645         | -11.14                | 845.39      | 1                      | 31.822564            | Keep Current Factory |
| 18 | Wonka Bar - Triple Dazzle Caramel | Pacific  | Wicked Choccy's | Lot's O' Nuts     | 1332.133136           | 1477.787346         | -10.93                | 878.52      | 1                      | 31.724430            | Keep Current Factory |
| 16 | Wonka Bar - Triple Dazzle Caramel | Gulf     | Wicked Choccy's | The Other Factory | 1328.669697           | 1516.000979         | -14.10                | 624.64      | 1                      | 31.540000            | Keep Current Factory |
| 10 | Wonka Bar - Milk Chocolate        | Pacific  | Wicked Choccy's | Lot's O' Nuts     | 1326.024024           | 1476.403496         | -11.34                | 878.52      | 1                      | 31.478430            | Keep Current Factory |



# CALCULATE OPTIMIZATION RPIS

# 1.1. Lead Time Reduction
avg_lead_time_reduction = simulation_01[
    simulation_01["Scenario Factory"] == simulation_01["Current Factory"]
][["Lead Time Reduction %"]].mean()

# 1.2. Profit Impact Stability
profit_stability = 1 -
    simulation_01["Profit Impact Stability"].mean() * 100

# 1.3. Scenario Confidence Score
# Based on model fit from network fit
scenario_confidence_score = 0.8389 * 100

# 1.4. Recommendation Coverage
total_product_region_pairs = len(rows_performance_reliable)

recommended_pairs = len(
    recommendation_table[
        recommendation_table["Recommendation"] == "Recommend Reassignment"
    ]
)

recommendation_coverage = (
    recommended_pairs / total_product_region_pairs
) * 100

print("OPTIMIZATION RPIS")
print("-" * 40)
print(f"/Average Lead Time Reduction: {avg_lead_time_reduction:.2f}%")
print(f"/Lead Time Reduction: {avg_lead_time_reduction:.2f}%")
print(f"/Profit Impact Stability: {profit_stability:.2f}%")
print(f"/Profit Stability: {profit_stability:.2f}%")
print(f"/Scenario Confidence Score: {scenario_confidence_score:.2f}%")
print(f"/Scenario Confidence Score: {scenario_confidence_score:.2f}%")
print(f"/Recommendation Coverage: {recommendation_coverage:.2f}%")
print(f"/Recommendation Coverage: {recommendation_coverage:.2f}%")

# 0.01

OPTIMIZATION RPIS
-----
Average Lead Time Reduction: -12.59%
```

[Route / Product-Region Analysis]

6. Key Results and Insights

- The dataset is structurally clean: duplicate rows and core validation errors were not detected.
- Order Year is a strong temporal signal: average lead time was approximately 1,094.02 for 2024 orders and 1,478.56 for 2025 orders.
- Gradient Boosting provided the strongest overall model according to RMSE and R^2 .
- 24 reliable product-region combinations were used for optimization rather than allowing low-volume combinations to dominate.
- 120 factory scenarios were evaluated across the five available factories.
- Zero alternative scenarios produced a predicted lead-time improvement.
- All 24 reliable product-region decisions therefore retain the current factory.
- Average predicted lead-time reduction across the evaluated scenarios was -12.59%, indicating that alternatives were generally predicted to worsen lead time.
- Profit impact stability was 100% under the project's stability assumption.
- The model-performance proxy used for scenario confidence is 53.38%, based on the Gradient Boosting R^2 .

```
# FINAL OPTIMIZATION SUMMARY

total_scenarios = len(simulation_df)

alternative_scenarios = simulation_df[
    simulation_df["Scenario Factory"] != simulation_df["Current Factory"]
]

beneficial_count = len(
    alternative_scenarios[
        alternative_scenarios["Lead Time Reduction %"] > 0
    ]
)

keep_current_count = len(
    recommendation_table[
        recommendation_table["Recommendation"] == "Keep Current Factory"
    ]
)

print("FINAL OPTIMIZATION SUMMARY")
print("-" * 45)

print(f"Total scenarios evaluated: {total_scenarios}")
print(f"Alternative factory scenarios: {len(alternative_scenarios)}")
print(f"Beneficial reassignment scenarios: {beneficial_count}")
print(f"Keep Current Factory decisions: {keep_current_count}")
print(f"Recommendation Coverage: {(recommendation_coverage:.1f)}%")

print("\nKEY FINDING")
print("-" * 45)

if beneficial_count == 0:
    print(
        "No alternative factory assignment produced a "
        "predicted lead-time improvement."
    )
    print(
        "The optimization system recommends retaining "
        "the current factory assignments."
    )
else:
    print(
        f"{beneficial_count} scenarios produced a "
        "predicted lead-time improvement."
    )

✓ 0.0s

=====
FINAL OPTIMIZATION SUMMARY
=====
Total scenarios evaluated: 120
Alternative factory scenarios: 96
Beneficial reassignment scenarios: 0
Keep Current Factory decisions: 24
Recommendation Coverage: 0.00%

KEY FINDING
-----
No alternative factory assignment produced a predicted lead-time improvement.
The optimization system recommends retaining the current factory assignments.
```

7. Discussion

The central finding is conservative but operationally useful: the system does not identify a supported reason to reassign products to alternative factories under the current model assumptions. This is a meaningful decision outcome because decision intelligence should be capable of recommending 'do not change' when the simulated alternatives are worse.

The result also highlights the importance of temporal information. The difference between 2024 and 2025 average lead times is substantially larger than the differences between regions. This suggests that the dataset contains time-dependent structure that should be monitored in future production versions.

The optimization framework should therefore be viewed as a repeatable scenario-testing mechanism rather than a one-time factory relocation tool.

8. Recommendations

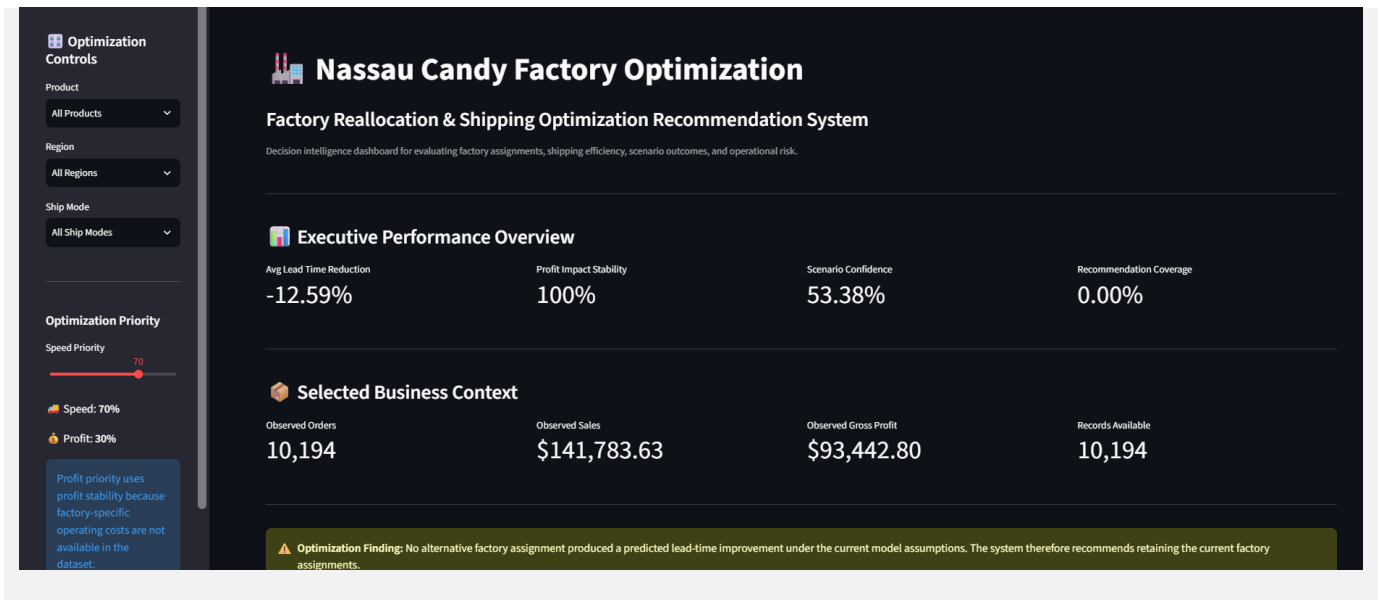
1. Retain current factory assignments under the current model and dataset assumptions.
2. Prioritize monitoring of high-volume chocolate products because improvements or deterioration in these products can affect the largest number of orders.
3. Re-run the scenario engine as new shipment data become available so that recommendations reflect current operating conditions.
4. Collect factory-specific operating costs, capacity, throughput and service-level data to enable real profit-impact optimization.
5. Replace approximate regional coordinates with actual customer or destination coordinates where available.
6. Monitor high-risk product-region combinations before changing factory assignments.
7. Use the Streamlit dashboard as the operational review layer for scenario comparison and decision documentation.

9. Streamlit Dashboard

The Streamlit application is the stakeholder-facing delivery layer. It converts the analytical workflow into an interactive interface where users can inspect factory scenarios, filter business context and review risk.

- Product selector
- Region selector
- Ship Mode context filter
- Speed-versus-profit optimization priority control
- Executive KPI overview
- Factory Optimization Simulator
- What-If Scenario Analysis
- Priority-based scenario ranking
- Recommendation Dashboard
- Risk & Impact Panel

- Executive Decision Summary
- Model and data assumptions documentation



[Dashboard]



[Factory Optimization Simulator]

What-If Scenario Analysis

Evaluate the predicted impact of assigning a product-region combination to different factories.

Select Product

Raisinables

Select Region

Atlantic

Predicted Lead-Time Change by Factory



Positive values indicate predicted improvement, negative values indicate predicted deterioration.

Scenario Factory	Predicted Lead Time	Lead Time Improvement	Lead Time Reduction %	Distance km	Profit Margin %	Profit Impact Stable
Sugar Shack	1450.7026	-177.5945	-13.95	1195.47	7.6923	☐
Wicked Choco's	1474.5956	-201.4875	-15.83	946.14	7.6923	☐
Let's O' Nuts	1481.6207	-207.9126	-16.33	3355.59	7.6923	☐
Secret Factory	1481.6207	-207.9126	-16.33	1347.26	7.6923	☐
The Other Factory	1498.5982	-225.4811	-17.71	1394.89	7.6923	☐

[What-If Scenario Analysis]

Priority-Based Scenario Ranking

The ranking below changes dynamically according to the Speed or Profit priority selected in the sidebar.

Product Name	Region	Current Factory	Scenario Factory	Lead Time Reduction %	Distance km	Profit Impact Stable	Scenario Score
Wanna Bar - Assorted Assortment	Interior	Let's O' Nuts	Sugar Shack	-6.14	1080.30	☐	86.807
Wanna Bar - Nutty Crunch Surprise	Interior	Let's O' Nuts	Sugar Shack	-6.77	1080.30	☐	86.5271
Wanna Bar - Nutty Crunch Surprise	Atlantic	Let's O' Nuts	Wicked Choco's	8.28	946.14	☐	85.2879
Lickable Whippersn	Atlantic	Secret Factory	Secret Factory	8.41	1347.26	☐	85.2861
Wanna Bar - Nutty Crunch Surprise	Atlantic	Let's O' Nuts	Secret Factory	8.96	1347.26	☐	83.2677
Wanna Bar - Nutty Crunch Surprise	Atlantic	Let's O' Nuts	Let's O' Nuts	-6.96	3355.59	☐	83.2677
Wanna Bar - Triple Devil Caramel	Interior	Wicked Choco's	Sugar Shack	-6.36	1080.30	☐	80.1484
Wanna Bar - Milk Chocolate	Interior	Wicked Choco's	Secret Factory	-6.41	845.36	☐	80.1526
Wanna Bar - Nutty Crunch Surprise	Atlantic	Let's O' Nuts	Secret Factory	8.96	845.36	☐	79.494
Wanna Bar - Triple Devil Caramel	Atlantic	Wicked Choco's	Sugar Shack	9.84	1085.49	☐	78.1812

Recommendation Dashboard

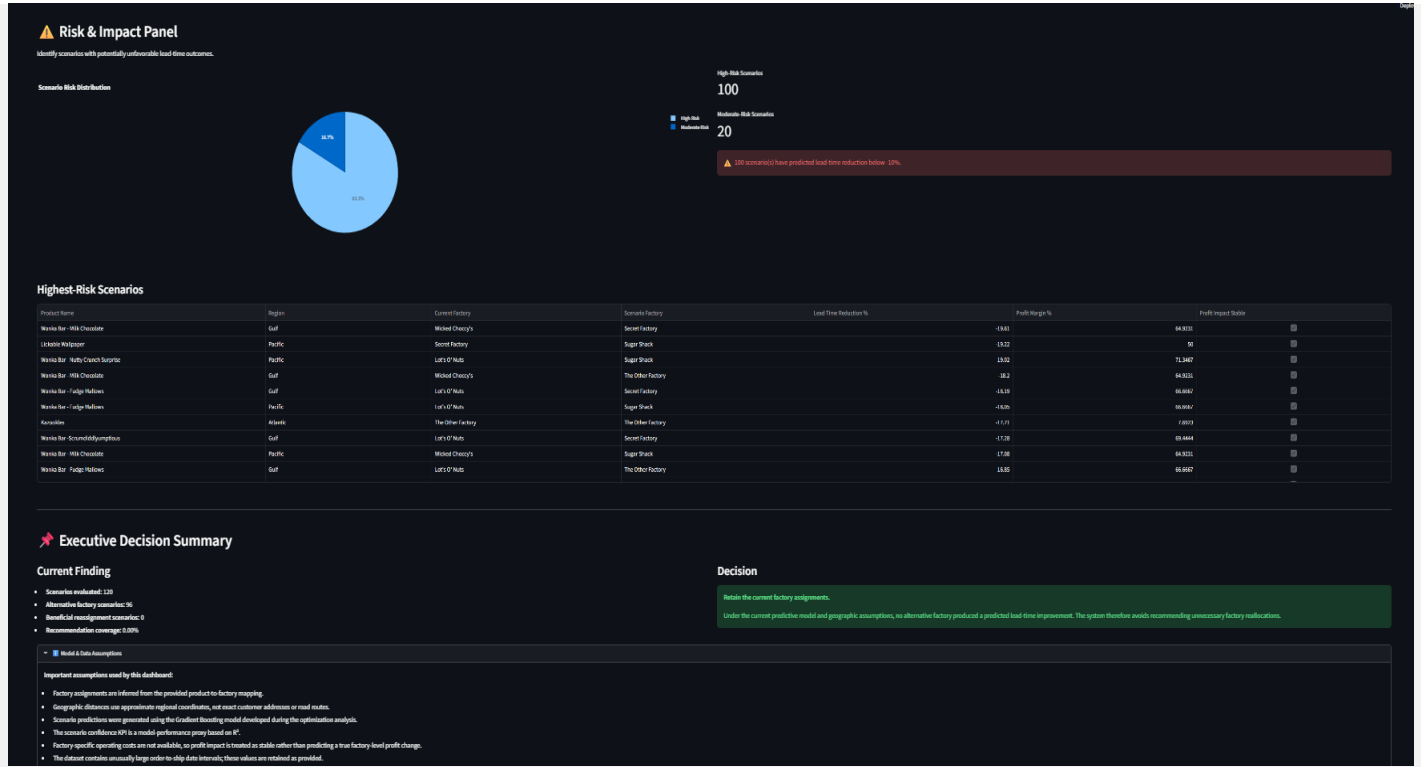
Factory decisions generated from the optimization engine.

Product Name	Region	Current Factory	Scenario Factory	Lead Time Reduction %	Distance km	Profit Stability Score	Recommendation Score	Recommendation	Status
Lickable Whippersn	Atlantic	Secret Factory	Wicked Choco's	-1.95	946.14	1	31.090	Keep Current Factory	Current Factory Retained
Wanna Bar - Milk Chocolate	Interior	Wicked Choco's	Secret Factory	9.41	946.14	1	32.696	Keep Current Factory	Current Factory Retained
Wanna Bar - Nutty Crunch Surprise	Atlantic	Let's O' Nuts	Wicked Choco's	8.39	946.14	1	32.761	Keep Current Factory	Current Factory Retained
Wanna Bar - Nutty Crunch Surprise	Interior	Let's O' Nuts	Secret Factory	8.59	945.39	1	32.768	Keep Current Factory	Current Factory Retained
Wanna Bar - Assorted Assortment	Interior	Let's O' Nuts	Secret Factory	8.29	945.39	1	30.688	Keep Current Factory	Current Factory Retained
Wanna Bar - Triple Devil Caramel	Interior	Wicked Choco's	Secret Factory	-16.15	945.39	1	31.086	Keep Current Factory	Current Factory Retained
Wanna Bar - Triple Devil Caramel	Interior	Let's O' Nuts	Secret Factory	-11.14	945.39	1	31.676	Keep Current Factory	Current Factory Retained
Wanna Bar - Triple Devil Caramel	Pacific	Wicked Choco's	Let's O' Nuts	-10.93	976.02	1	31.784	Keep Current Factory	Current Factory Retained
Wanna Bar - Triple Devil Caramel	Gulf	Wicked Choco's	The Other Factory	14.1	824.84	1	31.134	Keep Current Factory	Current Factory Retained
Wanna Bar - Milk Chocolate	Pacific	Wicked Choco's	Let's O' Nuts	11.34	976.02	1	31.434	Keep Current Factory	Current Factory Retained

Recommendation Distribution



[Recommendation Dashboard]



[Dashboard — Risk & Impact Panel]

10. Limitations and Future Work

- The dataset does not contain an explicit origin-factory field; factory assignment is inferred from the supplied product-to-factory mapping.
- The dataset contains unusually large order-to-ship date intervals. These values were retained rather than silently corrected.
- The geographic distance matrix uses approximate regional coordinates and therefore represents modeling assumptions rather than exact road distances.
- Factory-specific operating-cost data are unavailable, so the current system cannot calculate a true factory-level profit change.
- Scenario confidence is represented using model R² as a proxy and should not be interpreted as a probabilistic confidence interval.
- The current scenario simulation is aggregated at product-region level and uses the existing model setup for ship mode.
- Future versions should incorporate factory capacity, inventory, production throughput, transportation cost, actual route distance and service-level constraints.
- A future optimization layer could use mathematical programming once reliable cost, capacity and demand constraints are available.

11. Conclusion

This project demonstrates an end-to-end decision-intelligence workflow for factory allocation and shipping optimization. It combines data validation, feature engineering, predictive modeling, route clustering, geographic scenario simulation and recommendation logic in one analytical system.

The final optimization engine evaluated 120 factory scenarios and found no alternative assignment with a predicted lead-time improvement. The resulting recommendation to retain current factory assignments is therefore an evidence-based outcome rather than a forced positive recommendation.

The project also demonstrates product thinking through the Streamlit dashboard, where stakeholders can move from executive KPIs to individual product-region scenarios, recommendations and risk indicators.

Appendix A — Core KPIs

KPI	Observed Value	Interpretation
Average Lead Time Reduction	-12.59%	Alternatives are predicted to worsen lead time on average.
Profit Impact Stability	100%	All simulated scenarios remain within the project's stability threshold.
Scenario Confidence Score	53.38%	Model-performance proxy based on Gradient Boosting R^2 .
Recommendation Coverage	0.00%	No beneficial reassignment was identified.

Appendix B — Screenshot Checklist

Insert these screenshots into the marked placeholders in the paper:

8. 01 — Notebook 01 dataset loading and validation output.
9. 02 — Product Sales vs Gross Profit chart.
10. 03 — Product-region route analysis / reliable route output.
11. 04 — Model comparison: Linear Regression vs Random Forest vs Gradient Boosting.
12. 05 — Clustering silhouette scores / cluster summary.
13. 06 — Scenario simulation / 120 scenario output.
14. 07 — Final recommendation table from Notebook 02.
15. 08 — Streamlit dashboard top / Executive Performance Overview.
16. 09 — Streamlit Factory Optimization Simulator.
17. 10 — Streamlit What-If Scenario Analysis.
18. 11 — Streamlit Recommendation Dashboard.
19. 12 — Streamlit Risk & Impact Panel.

Appendix C — Project Deliverables

Deliverable	Purpose
01 — Data Validation Notebook	Analytical foundation and data quality validation
02 — Clustering & Optimization Notebook	Predictive modeling, clustering and scenario simulation
03 — Streamlit Preparation Notebook	Dashboard-ready datasets and filter preparation
Streamlit Dashboard	Interactive decision-support application
Research Paper	Detailed analytical and methodological narrative
Executive Summary	Short stakeholder-facing decision document